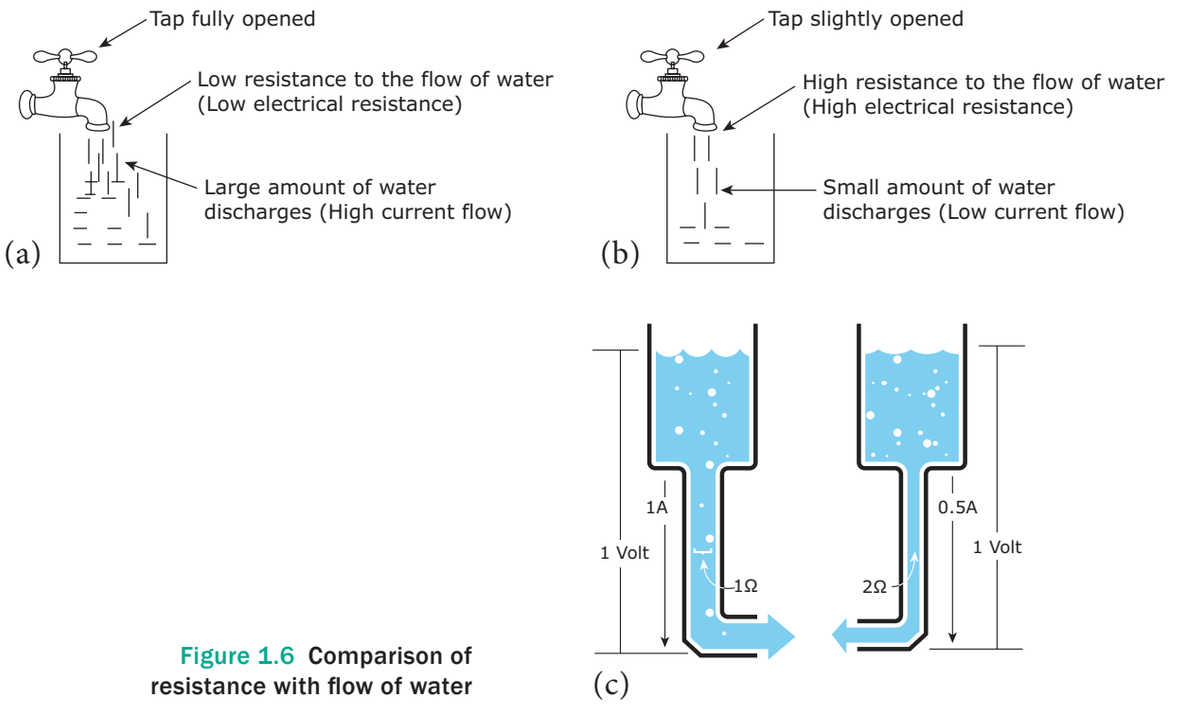


Figure 1.6 Comparison of resistance with flow of water

To understand "what is resistance", an analogy between the water-tap and the resistance is the simplest way. For example, if the tap is fully opened, maximum water flows through the tap, otherwise if a water tap is partially opened, then the amount of water flow through the tap is less.

Similarly, if the resistance offered to an electrical circuit is less, the amount of current flows in the circuit is maximum and vice versa.

In other words, the current depends upon the resistance offered by the circuit. This is shown in Figure 1.6(c).

1.3.3.1 Laws of resistance

The resistance of a conductor may vary according to the following factors.

1. It is directly proportional to the length of the conductor.
2. Inversely proportional to the cross sectional area of the conductor.
3. Directly proportional to the specific resistance or resistivity of the conductor.

$$R = \rho \frac{l}{A}$$

Where R is the resistance, l is the length, A is the area of cross-section, and ρ is the specific resistivity of the conductor as shown in Figure 1.7.

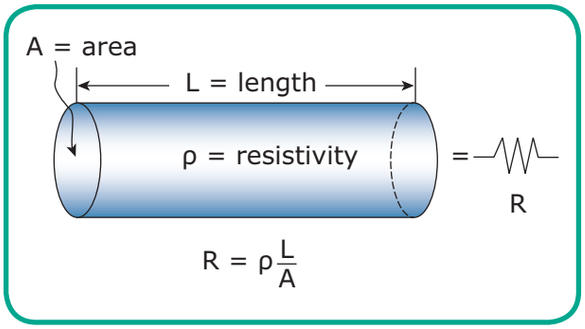


Figure 1.7 Factors affecting the resistance of a conductor

ELEC_EQUIP_Chapter01.indd 6 11/10/2020 12:22:37 PM